EEY4189 PROJECT PROPOSAL

Lost & Found Management System for Open University of Sri Lanka

Team Members: - S.W.D.K.M.A. Dinodi Rimesha Manamendra (Reg No: 232239) - F.Z. Fallil (Reg No: 422510475) - N.S.U.S. Kavinda (Reg No: 621435202)

---

1. INTRODUCTION AND BACKGROUND

The misplacement of personal belongings is a persistent challenge in educational institutions, particularly in large universities with dispersed campuses like the Open University of Sri Lanka. Students frequently lose items such as ID cards, mobile phones, wallets, USB drives, and textbooks across various locations including lecture halls, libraries, cafeterias, and common areas. The current informal methods of reporting lost items through notice boards, social media groups, or physical visits to security offices are inefficient and time-consuming.

Traditional lost and found systems rely heavily on manual processes where individuals must physically check multiple locations or rely on word-of-mouth communication. This approach lacks proper verification mechanisms, centralized information management, and real-time tracking capabilities. Security personnel often struggle to manage found items without a systematic way to catalog, store, and match them with their rightful owners. The absence of a digital platform results in prolonged item storage, increased risk of theft, and frustration for both students and staff.

Modern digital technologies offer an opportunity to transform this traditional approach into an efficient, transparent, and secure system. A dedicated Lost & Found Management System can provide a centralized platform where users can report, track, and claim their lost items with proper verification procedures. Unlike informal channels, this system implements features such as automated matching algorithms, notification alerts, secure verification processes, and detailed item categorization.

The proposed system addresses the specific needs of the Open University community while being scalable for future expansion. By digitizing the lost and found process, the university can significantly reduce the time required for item recovery, minimize administrative burden on security staff, and provide students with a reliable platform for retrieving their belongings. This initiative aligns with the university's digital transformation goals and demonstrates commitment to improving student services through technology adoption.

---

2. PROBLEM STATEMENT AND PROJECT OBJECTIVES

Problem Statement

The Open University of Sri Lanka lacks a centralized digital system for managing lost and found items, resulting in significant inefficiencies and frustrations for students, staff, and security personnel. Currently, students who lose personal belongings must rely on informal channels such as WhatsApp groups, physical notice boards, or visiting multiple security offices across different campus locations. This fragmented approach leads to several critical issues:

Students face difficulties in reporting lost items promptly and tracking their status, often resulting in permanent loss of valuable belongings. Security officers lack a systematic method to catalog found items, leading to accumulation of unclaimed items and storage challenges. There is no standardized verification process to ensure items are returned to legitimate owners, increasing the risk of fraudulent claims. The absence of real-time notifications means students may never learn that their lost items have been found. Additionally, the manual nature of current processes consumes significant time and resources from both students and administrative staff.

The problem is particularly acute given the university's distributed campus structure and large student population. Without a digital solution, the institution cannot provide adequate support for this common student need, potentially affecting student satisfaction and campus security.

Project Objectives

Primary Objective: To develop and implement a comprehensive web-based Lost & Found Management System that enables efficient reporting, tracking, and retrieval of lost items within the Open University of Sri Lanka premises, featuring secure verification processes and real-time notifications.

Secondary Objective: To enhance campus security and administrative efficiency by providing security officers with digital tools for cataloging suspicious or found items, managing handover processes, and generating reports on lost and found activities for better resource allocation and policy development.

---

3. INTRODUCTION TO SIMILAR SYSTEMS

Existing Lost and Found Systems Analysis

Several institutions and organizations have implemented digital lost and found systems with varying degrees of sophistication. **Crowdfind** is a popular platform used by universities in North America that allows students to report and search for lost items using detailed descriptions and location-based filtering. The

system includes photo uploads, automated matching algorithms, and email notifications when potential matches are found. However, it lacks robust verification mechanisms and relies primarily on user honesty.

RepoApp is another commercial solution deployed in airports and transportation hubs worldwide. It features QR code tagging for found items, multi-language support, and integration with security camera systems for location verification. The platform includes a mobile application for both iOS and Android, enabling users to report items on-the-go. Its strength lies in real-time tracking but requires significant infrastructure investment.

Campus Lost & Found by Indiana University provides a simpler approach focused specifically on educational institutions. The system integrates with student management systems for automatic user verification, implements a claim approval workflow involving administrative review, and generates monthly reports for campus security. While effective for its context, it lacks advanced features like image recognition or predictive analytics.

FindMyStuff, used by several corporate offices in Europe, emphasizes security and privacy. It implements blockchain technology for maintaining tamper-proof records of transactions, uses AI-powered image matching to identify similar items, and provides anonymous reporting options for sensitive items. The system's complexity, however, makes it less suitable for educational environments.

Our proposed system aims to combine the best features of these existing solutions while addressing their limitations. We plan to implement strong verification processes similar to Campus Lost & Found, incorporate photo-based reporting like Crowdfind, and maintain simplicity suitable for our university environment. Additionally, we will introduce unique features such as integration with student ID systems and customized workflows for different item categories.

---

4. PROPOSED SOLUTION

System Overview

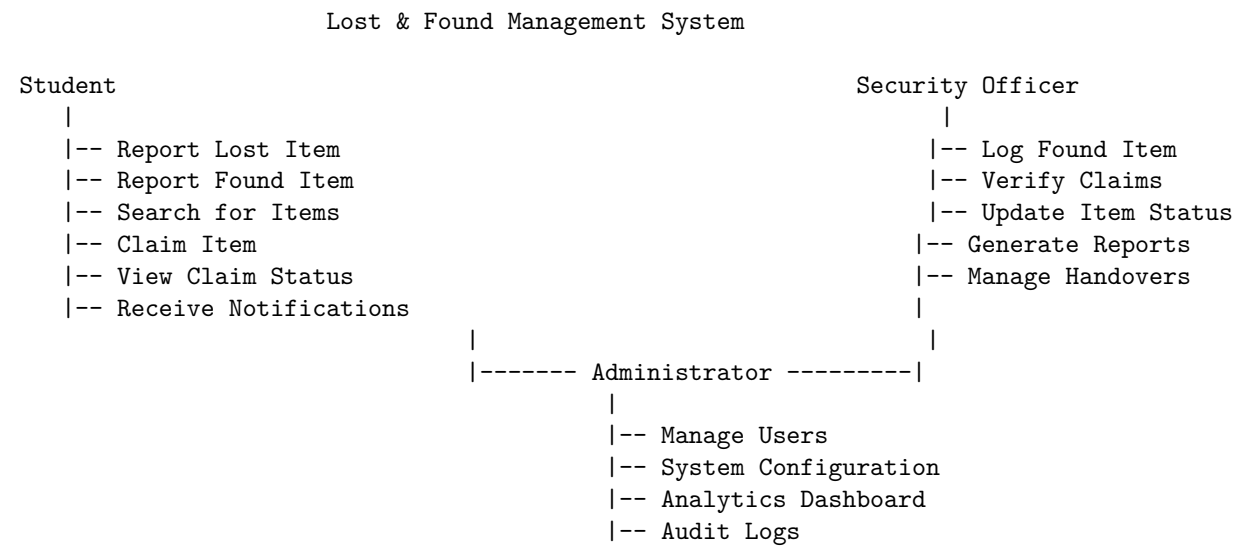
The proposed Lost & Found Management System is a comprehensive web-based platform designed to streamline the process of reporting, tracking, and retrieving lost items within the university. The system serves three primary user groups: students who report lost or found items, security officers who manage the physical items and verification processes, and administrators who oversee system operations and generate reports.

Key Features

The system implements a dual-reporting mechanism where both lost and found items can be registered with detailed descriptions including category, location, date/time, distinguishing features, and photographic evidence. An intelligent

matching algorithm automatically identifies potential matches between lost and found items based on multiple parameters. Users receive instant notifications via email and SMS when potential matches are identified. The verification process includes multi-factor authentication using student ID, security questions, and photo verification to ensure items are returned to rightful owners.

High-Level Use Case Diagram



System Workflow

When a student loses an item, they log into the system using university credentials and submit a detailed report. Simultaneously, when security finds an item, they create a found item entry. The system automatically cross-references new entries against existing records. When matches are identified, both parties receive notifications. Students can then initiate a claim process, providing additional verification details. Security officers review claims, verify ownership through the provided evidence, and approve legitimate claims. Finally, the system facilitates the physical handover process with proper documentation.

---

5. TECHNOLOGY STACK

Frontend Technologies

React.js 18.0 will serve as the primary frontend framework, chosen for its component-based architecture and extensive ecosystem. The user interface will be built using **Material-UI 5.0** for consistent, responsive design across devices. **Redux Toolkit** will manage application state, ensuring efficient data flow and

predictable state updates. For enhanced user experience, we'll implement **React Query** for server state management and caching. The frontend will be designed using **Figma** for wireframing and prototyping, with **Adobe XD** for creating interactive mockups.

Backend Technologies

The backend will be developed using **Node.js** with **Express.js** framework, providing a robust and scalable server environment. **MongoDB** will serve as the primary database, offering flexibility for storing varied item descriptions and user data. We'll implement **Mongoose ODM** for elegant MongoDB object modeling. **JWT (JSON Web Tokens)** will handle authentication and authorization. For real-time notifications, **Socket.io** will enable bidirectional communication between server and clients. **Redis** will be used for session management and caching frequently accessed data.

Development and Deployment Tools

Git with **GitHub** will manage version control and collaboration. **Docker** containers will ensure consistent development environments across team members. **Postman** will facilitate API testing and documentation. For database design, we'll use **MongoDB Compass** for visualization and **dbdiagram.io** for entity-relationship modeling. **Jest** and **React Testing Library** will handle unit and integration testing. The application will be deployed on **AWS EC2** instances with **Nginx** as a reverse proxy. **CloudFlare** will provide CDN services and DDoS protection.

Additional Technologies

Nodemailer will handle email notifications, integrated with the university's SMTP server. **Twilio API** will enable SMS notifications for urgent updates. **Multer** middleware will manage file uploads for item photos. **Sharp** library will handle image processing and optimization. **Winston** logger will maintain comprehensive application logs. **Helmet.js** and **express-rate-limit** will enhance security. For analytics, we'll implement **Google Analytics** and custom dashboard using **Chart.js**.

---

6. PROJECT TIMELINE AND CONCLUSION

Project Timeline (16 Weeks)

Phase 1: Planning and Design (Weeks 1-3) - Week 1: Requirements gathering, stakeholder interviews, and feasibility study - Week 2: System architecture design, database schema creation, and API specification - Week 3: UI/UX design, wireframe creation, and design approval

Phase 2: Development - Core Features (Weeks 4-8) - Weeks 4-5: Backend setup, database implementation, and authentication system - Weeks 6-7: Frontend development - user registration, login, and dashboard - Week 8: Implementation of lost/found item reporting features

Phase 3: Development - Advanced Features (Weeks 9-12) - Weeks 9-10: Matching algorithm development and notification system - Week 11: Claim verification workflow and security officer interface - Week 12: Admin panel and reporting features

Phase 4: Testing and Deployment (Weeks 13-15) - Week 13: Unit testing, integration testing, and bug fixes - Week 14: User acceptance testing with pilot group - Week 15: System deployment and documentation

Phase 5: Launch and Evaluation (Week 16) - Final system launch, user training, and post-implementation review

Conclusion

The proposed Lost & Found Management System addresses a critical need within the Open University of Sri Lanka by digitalizing and streamlining the process of reporting and retrieving lost items. By implementing modern web technologies and user-centric design principles, this system will significantly reduce the time and effort required for item recovery while enhancing security and accountability.

The project's success will be measured through metrics including reduction in permanently lost items, average retrieval time, user satisfaction scores, and administrative efficiency gains. Upon successful implementation, this system can serve as a model for other universities and potentially be scaled to serve multiple institutions.

This initiative represents a meaningful contribution to campus life improvement and demonstrates the practical application of software engineering principles to solve real-world problems.

Supervisor Approval:

Name: _____

Signature: _____

Date: _____